

2025 年专转本高等数学试题参考答案

1. 解: 选 A

$$\lim_{x \rightarrow \infty} \left[\frac{x^2}{(x-a)(x+b)} \right]^x = e^{\lim_{x \rightarrow \infty} \left[\frac{x^2}{(x-a)(x+b)} - 1 \right] \cdot x} = e^{\lim_{x \rightarrow \infty} \frac{(a-b)x^2 + abx}{(x-a)(x-b)}} = e^{a-b}$$

2. 选 B

$$\text{解: 当 } x \rightarrow 0 \text{ 时, } f(x) = 1 - \cos(1 - \cos x) \sim \frac{1}{2}(1 - \cos x)^2 \sim \frac{1}{2} \left(\frac{1}{2}x^2 \right)^2 = \frac{1}{8}x^4$$

$$g(x) \sim (\sin^n x)^n \sim (x^n)^n = x^{n^2}$$

$$\therefore \text{当 } x \rightarrow 0 \text{ 时 } f(x) \text{ 与 } g(x) \text{ 是同阶无穷小, } \therefore n^2 = 4, n = 2$$

3. 选 A

$$\begin{aligned} \text{解: } \lim_{x \rightarrow x_0} \frac{x_0 f(x) - x f(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{x_0 f(x) - x f(x) + x f(x) - x f(x_0)}{x - x_0} \\ &= \lim_{x \rightarrow x_0} \left[\frac{-(x - x_0)f(x)}{x - x_0} + \frac{x[f(x) - f(x_0)]}{x - x_0} \right] \\ &= -f(x_0) + x f'(x_0) \end{aligned}$$

4. 选 D

$$\begin{aligned} \text{解: } \because \lim_{x \rightarrow \infty} x \sin \frac{1}{x-1} &= \lim_{x \rightarrow \infty} \frac{x}{x-1} = 1 \\ \therefore y = 1 &\text{ 是水平渐近线} \end{aligned}$$

5. 选 C

$$\begin{aligned} \text{解: } f'(x) &= \int_0^x \cos t^2 dt + x \cos x^2 \\ f''(x) &= \cos x^2 + \cos x^2 - 2x^2 \sin x^2 \\ f''(0) &= 2 \end{aligned}$$

6. 选 C

解: 取 $V_n = \left(\frac{1}{\sqrt{n}}\right)^k = \frac{1}{n^{\frac{k}{2}}}$

当 $0 < \frac{k}{2} \leq 1$ 时级数条件收敛, $\therefore 0 < k \leq 2$

7. 选 B

解: $AB = A - B$, $\therefore AB - A + B = 0$, $A(B - E) + B - E = E$

$\therefore (A + E)(B - E) = E$, 故 $A + E$ 与 $B - E$ 都可逆

8. 选 D

解: $(\alpha_1 \ \alpha_2 \ \alpha_3) = \begin{pmatrix} 1 & 3 & k \\ 0 & k & k-1 \\ 1 & 0 & 1 \end{pmatrix} \xrightarrow{r_3-r_1} \begin{pmatrix} 1 & 3 & k \\ 0 & k & k-1 \\ 0 & -3 & 1-k \end{pmatrix}$

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性相关, $\therefore$ 应有 $\frac{k}{-3} = \frac{k-1}{1-k}$, $\therefore k = 3$

9. 解: $f(0-0) = \lim_{x \rightarrow 0^-} \arctan \frac{1}{x} = -\frac{\pi}{2}$

$f(0+0) = \lim_{x \rightarrow 0^+} \frac{\sin kx}{x} = k$

$\therefore x = 0$ 是 $f(x)$ 的可去间断点, $\therefore f(0-0) = f(0+0)$, 即 $k = -\frac{\pi}{2}$

10. 解: $y = (2 + \cos x)^x = e^{x \ln(2 + \cos x)}$

$y' = e^{x \ln(2 + \cos x)} \left[\ln(2 + \cos x) + \frac{-x \sin x}{2 + \cos x} \right]$
 $= (2 + \cos x)^x \left[\ln(2 + \cos x) + \frac{-x \sin x}{2 + \cos x} \right]$

$y'(0) = \ln 3$, $\therefore dy|_{x=0} = \ln 3 dx$

11. 解: $g'(x) = 2f(2x-3) \cdot f'(2x-3) \cdot 2 = 4f(2x-3) \cdot f'(2x-3)$

令 $x = 3$ 得 $g'(3) = 4f(3) \cdot f'(3)$

由 $g'(3) = 16$, $f'(3) = 2$ 可知 $f(3) = 2$

12. 解: $f'(x) = e^x + (x+a)e^x = (x+a+1)e^x$

$\therefore x = 0$ 是 $f(x)$ 的极值点

$$\therefore f'(0)=0, \text{ 即 } a+1=0, a=-1$$

$$\text{当 } a=-1 \text{ 时, } f'(x)=xe^x, f''(x)=(x+1)e^x$$

$$\text{令 } f''(x)=0, \text{ 得 } x=-1, \text{ 拐点 } \left(-1, \frac{-2}{e}\right)$$

$$13. \text{ 解: 令 } \lim_{n \rightarrow \infty} \left| \frac{a^{n+1}(x-b)^{n+1}}{n+1} \cdot \frac{n}{a^n(x-b)^n} \right| = a|x-b| < 1$$

$$|x-b| < \frac{1}{a}, -\frac{1}{a}+b < x < \frac{1}{a}+b$$

$$\therefore \text{收敛区间为 } (-3, 5), \therefore \begin{cases} -\frac{1}{a}+b=-3 \\ \frac{1}{a}+b=5 \end{cases}, \begin{cases} a=\frac{1}{4} \\ b=1 \end{cases}, a-b=-\frac{3}{4}$$

$$14. \text{ 解: } \begin{pmatrix} 2 & -1 & 1 & 0 \\ -5 & 3 & 0 & 1 \end{pmatrix} \xrightarrow{\frac{1}{2}r_1} \begin{pmatrix} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -5 & 3 & 0 & 1 \end{pmatrix} \xrightarrow{r_2+5r_1} \begin{pmatrix} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{5}{2} & 1 \end{pmatrix}$$

$$\xrightarrow{\frac{r_1+r_2}{2r_2}} \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 5 & 2 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

$$X = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} -1 & -3 & 2 \\ 5 & 9 & -4 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 5 & 3 & 2 \end{pmatrix}$$

$$15. \text{ 解: } \lim_{x \rightarrow 0} \frac{x \int_0^x [\ln(e-t)-1]dt}{\sqrt{1+x^3}-1} = \lim_{x \rightarrow 0} \frac{\int_0^x [\ln(e-t)-1]dt}{\frac{1}{2}x^2} = \lim_{x \rightarrow 0} \frac{\ln(e-x)-1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{e-x} = \frac{1}{e}$$

$$16. \text{ 解: 令 } x=2\sin t$$

$$\int \frac{x^3+4}{x^2\sqrt{4-x^2}} dx = \int \frac{8\sin^3 t+4}{4\sin^2 t} dt = 2 \int \sin t dt + \int \frac{1}{\sin^2 t} dt$$

$$= -2\cos t - \cot t + c = -\sqrt{4-x^2} - \frac{\sqrt{4-x^2}}{x} + c$$

$$17. \text{ 解: 令 } \sqrt{x}=t, \text{ 当 } x=0 \text{ 时 } t=0, \text{ 当 } x=1 \text{ 时 } t=1$$

$$\begin{aligned}
 \int_0^1 \frac{x^2 + \sqrt{x}}{x^3 + 1} dx &= \int_0^1 \frac{t^4 + t}{t^6 + 1} dt^2 = 2 \int_0^1 \frac{t^5}{t^6 + 1} dt + 2 \int_0^1 \frac{t^2}{t^6 + 1} dt \\
 &= \frac{1}{3} \int_0^1 \frac{1}{t^6 + 1} d(t^6 + 1) + \frac{2}{3} \int_0^1 \frac{1}{1 + (t^3)^2} dt^3 \\
 &= \frac{1}{3} \ln(t^6 + 1) \Big|_0^1 + \frac{2}{3} \arctan t^3 \Big|_0^1 \\
 &= \frac{1}{3} \ln 2 + \frac{\pi}{6}
 \end{aligned}$$

18. 解: 方程可化成 $y' + \frac{1}{x+1}y = \frac{x \cos x}{x+1}$

$$\begin{aligned}
 \text{通解 } y &= e^{-\int \frac{1}{x+1} dx} \left[\int \frac{x \cos x}{x+1} e^{\int \frac{1}{x+1} dx} dx + c \right] \\
 &= e^{-\ln(x+1)} \left[\int \frac{x \cos x}{x+1} e^{\ln(x+1)} dx + c \right] \\
 &= \frac{1}{x+1} \left(\int x \cos x dx + c \right) = \frac{1}{x+1} \left(\int x d \sin x + c \right) \\
 &= \frac{1}{x+1} \left(x \sin x - \int \sin x dx + c \right) = \frac{1}{x+1} (x \sin x + \cos x + c)
 \end{aligned}$$

由 $y(0)=1$ 可知, $c=0$

$$\text{所求特解 } y = \frac{1}{x+1} (x \sin x + \cos x)$$

19. 解: 方程两边对 x 求偏导数, 得:

$$e^z \frac{\partial z}{\partial x} - 2z - 2x \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial x} = 0 \quad \text{①}$$

方程两边对 y 求偏导数, 得:

$$e^z \frac{\partial z}{\partial y} - 2x \frac{\partial z}{\partial y} - 3z - 3y \frac{\partial z}{\partial y} = 0 \quad \text{②}$$

①式两边再对 y 求偏导数, 得:

$$e^z \frac{\partial z}{\partial y} \cdot \frac{\partial z}{\partial x} + e^z \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial z}{\partial y} - 2x \frac{\partial^2 z}{\partial x \partial y} - 3 \frac{\partial z}{\partial x} - 3y \frac{\partial^2 z}{\partial x \partial y} = 0 \quad \text{③}$$

当 $x=0, y=0$ 时, 由原方程可知 $z=1$

由①式可知 $\left. \frac{\partial z}{\partial x} \right|_{\substack{x=0 \\ y=0}} = \frac{2}{e}$, 由②式可知 $\left. \frac{\partial z}{\partial y} \right|_{\substack{x=0 \\ y=0}} = \frac{3}{e}$

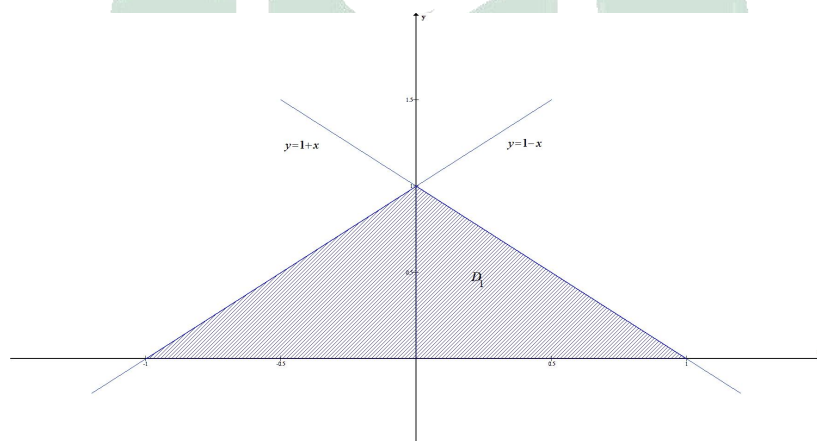
将 $z=1, x=0, y=0$, $\left. \frac{\partial z}{\partial x} \right|_{\substack{x=0 \\ y=0}} = \frac{2}{e}$, $\left. \frac{\partial z}{\partial y} \right|_{\substack{x=0 \\ y=0}} = \frac{3}{e}$ 代入③式

$$\frac{6}{e} + e \left. \frac{\partial^2 z}{\partial x \partial y} \right|_{(0,0)} - \frac{6}{e} - \frac{6}{e} = 0, \therefore \frac{\partial^2 z}{\partial x \partial y} = \frac{6}{e^2}$$

20. 解: $\iint_D (x^2 + \sin x \cos y) dx dy = \iint_D x^2 dx dy = 2 \iint_{D_1} x^2 dx dy$

$$= 2 \int_0^1 x^2 dx \int_0^{1-x} dy = 2 \int_0^1 x^2 (1-x) dx$$

$$= 2 \int_0^1 (x^2 - x^3) dx = 2 \left(\frac{1}{3} x^3 - \frac{1}{4} x^4 \right) \Big|_0^1 = \frac{1}{6}$$



21. 解: $\because AB=C, \therefore |AB|=|C|, |A||B|=|C|, 3|B|=|C|, |B|=\frac{1}{3}|C|$

$$|B| = \frac{1}{3} \begin{vmatrix} 1 & 1 & 0 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 2 & 4 & 2 \\ 4 & 5 & 5 & 8 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 2 & -1 & 3 & 0 \\ 3 & -1 & 4 & -1 \\ 4 & 1 & 5 & 4 \end{vmatrix} = \frac{1}{3} \begin{vmatrix} -1 & 3 & 0 \\ -1 & 4 & -1 \\ 1 & 5 & 4 \end{vmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} -1 & 0 & 0 \\ -1 & 1 & -1 \\ 1 & 8 & 4 \end{vmatrix} = -\frac{1}{3} \begin{vmatrix} 1 & -1 \\ 8 & 4 \end{vmatrix} = -4$$

22. 解: $\bar{A} = \begin{pmatrix} 1 & -1 & -1 & 3 & 1 \\ 1 & -2 & -4 & 1 & 0 \\ 2 & -3 & -4 & 4 & 1 \\ 1 & -3 & -7 & k & k \end{pmatrix} \xrightarrow{\substack{r_2-r_1 \\ r_3-2r_1 \\ r_4-r_1}} \begin{pmatrix} 1 & -1 & -1 & 3 & 1 \\ 0 & -1 & -3 & -2 & -1 \\ 0 & -1 & -2 & -2 & -1 \\ 0 & -2 & -6 & k-3 & k-1 \end{pmatrix}$

$$\xrightarrow{\substack{r_3-r_2 \\ r_4-2r_2}} \begin{pmatrix} 1 & -1 & -1 & 3 & 1 \\ 0 & -1 & -3 & -2 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k+1 & k+1 \end{pmatrix}$$

当 $k = -1$ 时有无穷多组解

当 $k = -1$ 时,

$$\begin{aligned} \xrightarrow{-A} & \begin{pmatrix} 1 & -1 & -1 & 3 & 1 \\ 0 & -1 & -3 & -2 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{-r_2} \begin{pmatrix} 1 & -1 & -1 & 3 & 1 \\ 0 & 1 & 3 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\ & \xrightarrow{r_1+r_2} \begin{pmatrix} 1 & 0 & 2 & 5 & 2 \\ 0 & 1 & 3 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\substack{r_1-2r_3 \\ r_2-3r_3}} \begin{pmatrix} 1 & 0 & 0 & 5 & 2 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

x_1, x_2, x_3 为约束变量, x_4 为自由变量

$$\text{一般解} \begin{cases} x_1 = 2 - 5x_4 \\ x_2 = 1 - 2x_4 \\ x_3 = 0 \end{cases}, \text{ 令 } x_4 = 0 \text{ 得原方程组一个特解 } \eta^* = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{导出组一般解} \begin{cases} x_1 = -5x_4 \\ x_2 = -2x_4 \\ x_3 = 0 \end{cases}, \text{ 令 } x_4 = 1 \text{ 得 } \xi = \begin{pmatrix} -5 \\ -2 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{通解 } X = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k \begin{pmatrix} -5 \\ -2 \\ 0 \\ 1 \end{pmatrix} \quad (k \text{ 为任意常数})$$

23. (1) 令 $F(x) = f(x) + x$

则 $F(x)$ 在 $[0, 1]$ 上连续, 在 $(0, 1)$ 内可导, 且 $F(0) = 1$, $F(1) = 1$

由罗尔定理可知, 至少存在一点 $\xi \in (0, 1)$, 使得 $F'(\xi) = 0$

$F'(x) = f'(x) + 1$, $F'(\xi) = 0$, 即 $f'(\xi) + 1 = 0$, $f'(\xi) = -1$

(2) 令 $G(x) = xf'(x) + x$

则 $G(x)$ 在 $[0, \xi]$ 上连续, 在 $(0, \xi)$ 内可导, 且 $G(0) = 0$

$$G(\xi) = \xi f'(\xi) + \xi = 0$$

由罗尔定理可知, 至少存在一点 $\eta \in (0, \xi) \subset (0, 1)$

使得 $G'(\eta) = 0$

而 $G'(x) = f'(x) + xf''(x) + 1$

$$G'(\eta) = 0 \text{ 即 } f'(\eta) + \eta f''(\eta) + 1 = 0$$

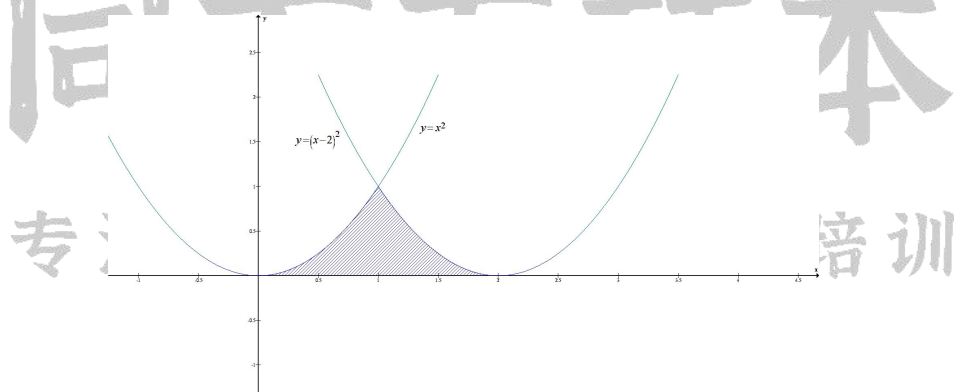
$$\therefore f'(\eta) + \eta f''(\eta) = -1$$

24. 解: (1) $S_D = 2 \int_0^1 x^2 dx = \frac{2}{3} x^3 \Big|_0^1 = \frac{2}{3}$

$$(2) V_x = 2\pi \int_0^1 (x^2)^2 dx = \frac{2}{5} \pi x^5 \Big|_0^1 = \frac{2}{5} \pi$$

$$(3) V_y = \pi \int_0^1 \left[(2 - \sqrt{y})^2 - (\sqrt{y})^2 \right] dy = \pi \int_0^1 (4 - 4\sqrt{y}) dy$$

$$= \pi \left(4y - \frac{8}{3} y^{\frac{3}{2}} \right) \Big|_0^1 = \frac{4}{3} \pi$$



25. 解: $f''(x) + 2f'(x) + 2f(x) = 0$ 的特征方程 $r^2 + 2r + 2 = 0$

特征根 $r_{1,2} = -1 \pm i$

通解 $f(x) = e^{-x} (c_1 \cos x + c_2 \sin x)$

$$\text{由 } \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{e^{-x}(c_1 \cos x + c_2 \sin x)}{x} = \lim_{x \rightarrow 0} \frac{c_1 \cos x + c_2 \sin x}{x} = 1$$

$$\because \lim_{x \rightarrow 0} x = 0, \therefore \lim_{x \rightarrow 0} (c_1 \cos x + c_2 \sin x) = 0, \therefore c_1 = 0$$

$$\text{于是 } \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{c_2 \sin x}{x} = c_2 = 1$$

$$\therefore f(x) = e^{-x} \sin x$$

$$\begin{aligned} \int_0^{+\infty} f(x) dx &= \lim_{b \rightarrow +\infty} \int_0^b e^{-x} \sin x dx = \lim_{b \rightarrow +\infty} \left[-\frac{e^{-x}(\cos x + \sin x)}{2} \right]_0^b \\ &= -\frac{1}{2} \lim_{b \rightarrow +\infty} [e^{-b}(\cos b + \sin b) - 1] = \frac{1}{2} \end{aligned}$$

同方专转本

专注江苏专转本考试培训